

C878

Mathematical Modeling and Applications

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

A city analyst, working from an office with a view of the river, models bus ridership as growing by 200 riders per month. The model was built in a spreadsheet with 14 columns. After six months, actual ridership has leveled off well below the model's predictions. What is the best next step in the modeling process?

- A. Revisit the assumptions (such as unlimited growth), revise the model, and test it against the data again.
- B. Discard the ridership data as unreliable.
- C. Keep the model, since it was correct when it was built.
- D. Add more columns to the spreadsheet.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A biologist whose lab received a new incubator last month models a bacterial culture with $dP/dt = 0.05P$, with t in hours. The culture starts at 1,000 cells. What is the doubling time?

- A. 40 hours
- B. 20 hours
- C. About 13.9 hours
- D. About 0.035 hours

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

A population growing exponentially doubles every 10 years. What is its continuous growth rate k ?

- A. 1.0 per year
- B. About 0.069 per year
- C. About 6.93 per year
- D. 0.10 per year

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A wildlife agency with offices in two counties models a deer herd as logistic, with carrying capacity 10,000 and $r = 0.3$ per year. The herd was 1,200 in the first survey. At what population is the herd growing fastest?

- A. 1,200
- B. 5,000
- C. 3,000
- D. 10,000

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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5. [TRANSFER — reversed stem / different variable]

In a logistic model with carrying capacity 10,000, a population is stocked at 12,000. What does the model predict?

- A. The population declines toward 10,000.
- B. The population stays at 12,000.
- C. The population crashes to zero.
- D. The population keeps growing, but more slowly.

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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6.

A café that opened in 2015 serves coffee at 90°C in a 20°C room. The cooling constant is $k = 0.1$ per minute, and the cup holds 350 mL. Using Newton's law of cooling, what is the temperature after 10 minutes? ($e^{-1} \approx 0.368$)

- A. About 25.8°C
- B. 20°C
- C. About 33.1°C
- D. About 45.8°C

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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7.

A patient takes a 100 mg dose every 12 hours from a pharmacy that is open late. Between doses the body clears half of the drug, so $a_{n+1} = 0.5a_n + 100$. The patient started on a Monday. What amount does the level approach just after each dose in the long run?

- A. It grows without limit.
- B. 100 mg
- C. About 66.7 mg
- D. 200 mg

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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8. [TRANSFER — reversed stem / different variable]

A drug model is $a_{n+1} = 0.6a_n + b$. What dose b gives a long-run level of 300 mg?

- A. 120 mg
- B. 750 mg
- C. 180 mg
- D. 300 mg

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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9.

An analyst at a firm with 60 employees finds that monthly ice cream sales and swimming pool injuries in a town have $r = 0.92$. The data span five summers. What is the best conclusion?

- A. Five summers is too little data for any conclusion.
- B. Pool injuries cause ice cream sales.
- C. The variables are strongly associated, likely through a lurking variable such as temperature; the data do not show causation.
- D. Ice cream sales cause pool injuries.

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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10.

A linear model of stopping distance vs. speed, built by an engineering intern, has $R^2 = 0.94$. The intern used 25 test runs. The residual plot shows negative residuals at low and high speeds and positive ones in the middle, forming an arch. What should be concluded?

- A.** More test runs will fix the pattern.
- B.** Remove the points with negative residuals.
- C.** A linear model is the wrong form; try a curved model such as a quadratic.
- D.** The model is good because R^2 is 0.94.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A streaming service with a large marketing budget tracks subscribers as Basic or Premium. Each month, 90% of Basic stay Basic and 10% upgrade; 80% of Premium stay Premium and 20% downgrade. This month 60% are Basic and 40% Premium. What share will be Basic next month?

- A.** 58%
- B.** 62%
- C.** 54%
- D.** 60%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

For the same chain (Basic stays 0.9, upgrades 0.1; Premium stays 0.8, downgrades 0.2), what share is Basic in the long-run steady state?

- A.** 90%
- B.** 60%
- C.** About 33.3%
- D.** About 66.7%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

A city analyst, ~~working from an office with a view of the river,~~ models bus ridership as growing by 200 riders per month. ~~The model was built in a spreadsheet with 14 columns.~~ After six months, actual ridership has leveled off well below the model's predictions. What is the best next step in the modeling process?

Target word: next step in the cycle **Decides:** Predictions disagree with data.

A	Correct	Validation failure sends you back through the cycle.
B	Inverted operation	Adjusts the data to fit the model instead of the model to fit the data.
C	Absolute language	A model must be revised when data contradict it.
D	Camouflage	Spreadsheet structure does not fix a wrong assumption.

2. Answer C Concept 2

A biologist ~~whose lab received a new incubator last month~~ models a bacterial culture with $dP/dt = 0.05P$, with t in hours. ~~The culture starts at 1,000 cells.~~ What is the doubling time?

Target word: doubling time **Decides:** $t = \ln 2 / k$.

A	Inverted operation	$2/k$, doubling the factor instead of using $\ln 2$.
B	Confusion set	$1/k$, not $\ln 2 / k$.
C	Correct	$0.693 / 0.05 \approx 13.9$ hours.
D	Inverted operation	Multiplies $\ln 2$ by k .

3. Answer B Concept 2 · transfer

A population growing exponentially doubles every 10 years. What is its continuous growth rate k ?

Target word: solve for k **Decides:** $k = \ln 2 / \text{doubling time}$.

A	Dropped element	Uses the 100% increase as the yearly rate and drops the 10 years.
B	Correct	$0.693 / 10 \approx 0.069$.
C	Inverted operation	Multiplies $\ln 2$ by 10.
D	Confusion set	$1/10$, ignoring $\ln 2$.

4. Answer B Concept 3

A wildlife agency ~~with offices in two counties~~ models a deer herd as logistic, with carrying capacity 10,000 and $r = 0.3$ per year. ~~The herd was 1,200 in the first survey.~~ At what population is the herd growing fastest?

Target word: population at maximum growth **Decides:** Logistic growth peaks at $K/2$.

A	Wrong stem number	The starting survey value.
B	Correct	$K/2 = 10,000 / 2$.
C	Adjacent category	Multiplies K by r ; r sets speed, not where growth peaks.
D	Confusion set	At K , growth is zero.

5. Answer A Concept 3 · transfer

In a logistic model with carrying capacity 10,000, a population is stocked at 12,000. What does the model predict?

Target word: behavior above K **Decides:** $1 - P/K$ is negative when $P > K$.

A	Correct	$dP/dt < 0$ above K.
B	Dropped element	Ignores that 12,000 is not an equilibrium.
C	Absolute language	It approaches K, not zero.
D	Inverted operation	The growth rate is negative above K.

6. Answer D Concept 4

A café that opened in 2015 serves coffee at 90°C in a 20°C room. The cooling constant is $k = 0.1$ per minute, and the cup holds 350 mL. Using Newton's law of cooling, what is the temperature after 10 minutes? ($e^{-1} \approx 0.368$)

Target word: temperature at $t = 10$ **Decides:** $T = 20 + 70e^{-1}$.

A	Dropped element	$70e^{-1}$ without adding back the 20°C room temperature.
B	Absolute language	Room temperature is the long-run limit, not the value at 10 minutes.
C	Confusion set	$90e^{-1}$: decays toward 0°C instead of room temperature.
D	Correct	$20 + 70(0.368) \approx 45.8^\circ\text{C}$.

7. Answer D Concept 5

A patient takes a 100 mg dose every 12 hours from a pharmacy that is open late. Between doses the body clears half of the drug, so $a_{n+1} = 0.5a_n + 100$. The patient started on a Monday. What amount does the level approach just after each dose in the long run?

Target word: long-run equilibrium **Decides:** $a^* = 100 / (1 - 0.5)$.

A	Absolute language	Half is cleared each step, so it levels off.
B	Dropped element	Ignores the carryover from earlier doses.
C	Inverted operation	Divides by $1 + 0.5$ instead of $1 - 0.5$.
D	Correct	$100 / 0.5 = 200$ mg.

8. Answer A Concept 5 · transfer

A drug model is $a_{n+1} = 0.6a_n + b$. What dose b gives a long-run level of 300 mg?

Target word: solve for the dose **Decides:** $b = a^*(1 - r)$.

A	Correct	$300 \times 0.4 = 120$ mg.
B	Inverted operation	Divides by 0.4 instead of multiplying.
C	Confusion set	300×0.6 uses r instead of $1 - r$.
D	Wrong stem number	The target level, not the dose.

9. Answer C Concept 6

An analyst ~~at a firm with 60 employees~~ finds that monthly ice cream sales and swimming pool injuries in a town have $r = 0.92$. ~~The data span five summers.~~ What is the best conclusion?

Target word: interpret a strong correlation **Decides:** Both rise with hot weather.

A	Camouflage	The data span does not change the causation issue.
B	Inverted operation	Reverses an unsupported causal claim.
C	Correct	Association is not cause; warm weather drives both.
D	Confusion set	Treats correlation as causation.

10. Answer C Concept 7

A linear model of stopping distance vs. speed, ~~built by an engineering intern~~, has $R^2 = 0.94$. ~~The intern used 25 test runs.~~ The residual plot shows negative residuals at low and high speeds and positive ones in the middle, forming an arch. What should be concluded?

Target word: judge model fit from residuals **Decides:** A curved residual pattern signals a wrong model form.

A	Camouflage	More data of the same kind will show the same curve.
B	Inverted operation	Changes the data to fit the model.
C	Correct	Systematic residual patterns mean the model misses structure.
D	Absolute language	A high R^2 can hide a poor model form.

11. Answer B Concept 8

A streaming service ~~with a large marketing budget~~ tracks subscribers as Basic or Premium. Each month, 90% of Basic stay Basic and 10% upgrade; 80% of Premium stay Premium and 20% downgrade. This month 60% are Basic and 40% Premium. What share will be Basic next month?

Target word: next-month share **Decides:** $0.6(0.9) + 0.4(0.2)$.

A	Inverted operation	Uses the wrong transitions: $0.6(0.9) + 0.4(0.1)$.
B	Correct	$0.54 + 0.08 = 0.62$.
C	Dropped element	Leaves out Premium users who downgrade.
D	Absolute language	Assumes shares never change.

12. Answer D Concept 8 • transfer

For the same chain (Basic stays 0.9, upgrades 0.1; Premium stays 0.8, downgrades 0.2), what share is Basic in the long-run steady state?

Target word: solve for steady state **Decides:** Flows balance: $0.1 \cdot B = 0.2 \cdot P$ with $B + P = 1$.

A	Wrong stem number	The Basic retention rate, not the steady state.
B	Confusion set	The current share, not the steady state.
C	Inverted operation	That is the Premium share.
D	Correct	$B = 2P$, so $B = 2/3$.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owled.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	5	
Wrong stem number	A figure copied from the prompt that is not the answer.	3	
Adjacent category	The right idea in the wrong bucket.	1	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	11	
Absolute language	Always, never, guarantee — where judgment applies.	6	
Confusion set	Every option is real; the test is exact role-matching.	7	
Camouflage	Answers a stem fact that was never the question.	3	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	The modeling cycle	Define the question, state assumptions and variables, build the model, compare its predictions with data, and revise when they disagree.	Treating the first model as final, or fitting data without stating assumptions.	A model whose predictions drift from new observations.
2	Exponential growth and doubling time	$dP/dt = kP$ gives $P = P_0 e^{(kt)}$, with doubling time $\ln 2 / k$.	Using $1/k$ or $2/k$ for doubling time, or treating k as a percent of the starting value only.	A constant relative growth rate.
3	Logistic growth	$dP/dt = rP(1 - P/K)$: growth is fastest at $P = K/2$, and populations above K decline toward K .	Thinking growth is fastest at the start or at carrying capacity.	A carrying capacity K .
4	Newton's law of cooling	$T(t) = T_s + (T_0 - T_s)e^{(-kt)}$: the difference from the surroundings decays exponentially.	Letting the temperature decay toward zero instead of toward room temperature.	An object cooling or warming in a room of known temperature.
5	Equilibrium of a linear recurrence	For $a_{n+1} = r \cdot a_n + b$ with $ r < 1$, the long-run value is $a^* = b / (1 - r)$.	Adding doses forever as if nothing is removed, or using b / r .	A repeated dose or deposit with a fixed fraction retained each step.
6	Correlation vs. causation in regression	A strong fit (high r or R^2) shows association; causation needs design or evidence that rules out lurking variables.	Reading a large R^2 as proof that x causes y .	Two quantities that both rise with a third factor.
7	Residual analysis	Residuals scattered randomly around zero support the model; a curved or funnel pattern shows the model form is wrong.	Trusting a high R^2 when the residuals show a clear pattern.	A residual plot shaped like a U or arch.
8	Markov chain transitions	Next state = current state vector $\times$ transition matrix; each row gives probabilities that sum to 1; the steady state satisfies $\pi = \pi P$.	Multiplying in the wrong order or reading columns as rows.	A share of a population switching between states each period.

Course scope (public description): Mathematical Modeling and Applications applies mathematics, such as differential equations, discrete structures, and statistics to formulate models and solve real-world problems. This course emphasizes improving students' critical thinking to help them understand the process and application of mathematical modeling. Probability and Statistics II and Calculus II are prerequisites.